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Brightlearn

Exercise 1

SQL Fundamentals

Section 1 : Selecting & Sorting

Question 1

- SELECT *
FROM employees;

employees						
ID	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesb
2	Brian	Lee	HR	45000	2019-02-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesb

Question 2

SELECT DISTINCT department
FROM employees;

Departments
IT
HR
Finance
Marketing

Question 3

```
SELECT first_name,  
       last_name  
FROM   employees  
ORDER BY salary DESC;
```

first - name	last - name
Eva	Naidoo
Alice	Green

Question 4

```
SELECT id,  
       first_name,  
       last_name,  
       salary  
FROM   employees  
ORDER BY salary DESC  
LIMIT 3;
```

id	first - name	last - name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Section 2: Filtering with WHERE

Question 5

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Question 6

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary  
FROM employees  
WHERE department = 'Finance' AND salary > 6000;
```

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

Question 7

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM employees  
WHERE department = 'HR' OR 'Marketing';
```

id	first-name	last-name	department
4	David	Mokoena	Marketing

Question 8

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM employees  
WHERE department NOT 'IT';
```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zule	Finance
4	David	Mokoena	Marketing

Question 9

```
SELECT id,  
       first - name,  
       last - name,  
       department  
FROM   employees  
WHERE  department IN ('IT', 'HR') OR ('Finance');
```

id	first - name	last - name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT
2	Brian	Lee	HR

Question 10

```
SELECT id,  
       first - name,  
       last - name,  
       department,  
       salary,  
       city  
FROM   employees  
WHERE  department = 'IT' salary > 65000 AND city = 'Johannesburg';
```

ID	first - name	last - name	department	Salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg